

Individual Design Report: Part Design for Manufacture

Part 1: 3D printed

Final 3D printed product:

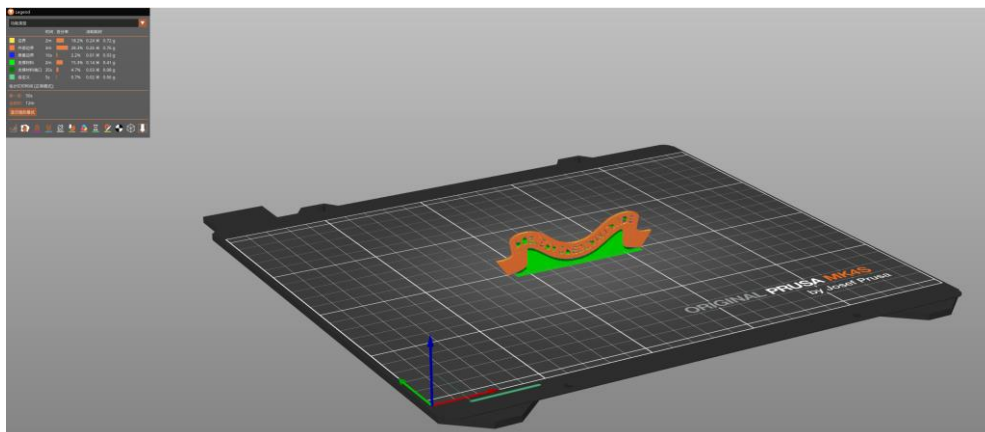


3D printed product 1



3D printed product 2

Slicing Software Output: Prints an overview of time and stratification data:



Visual preview of support structure in PrusaSlicer

打印设置:

  0.20mm SPEED (修改)  

耗材:

  Prusa PLA  

打印机:

  Original Prusa MK4S 0.4 nozzle  

支持: 任何地方 

填充: 15%  裙边: ☐

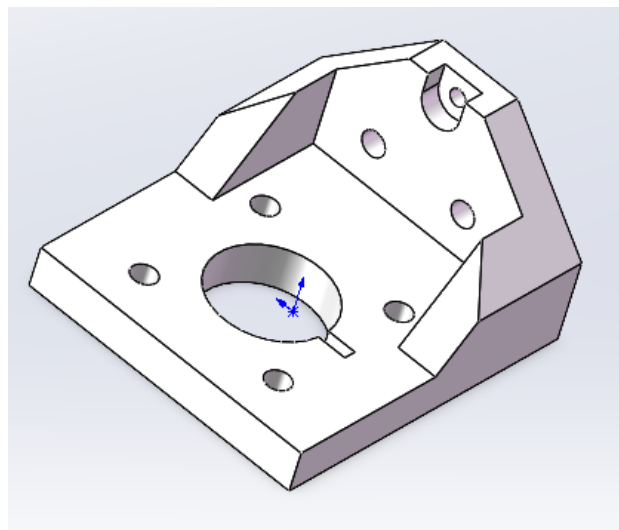
Prusa MK4S-based print setup parameter display

切片信息	
消耗耗材 (g) (包括线轴)	2.06 (232.06)
消耗耗材 (m)	0.69
消耗耗材 (mm ³)	1657.70
费用	0.06
预计打印时间:	
- 正常模式	12m
- 隐身模式	14m

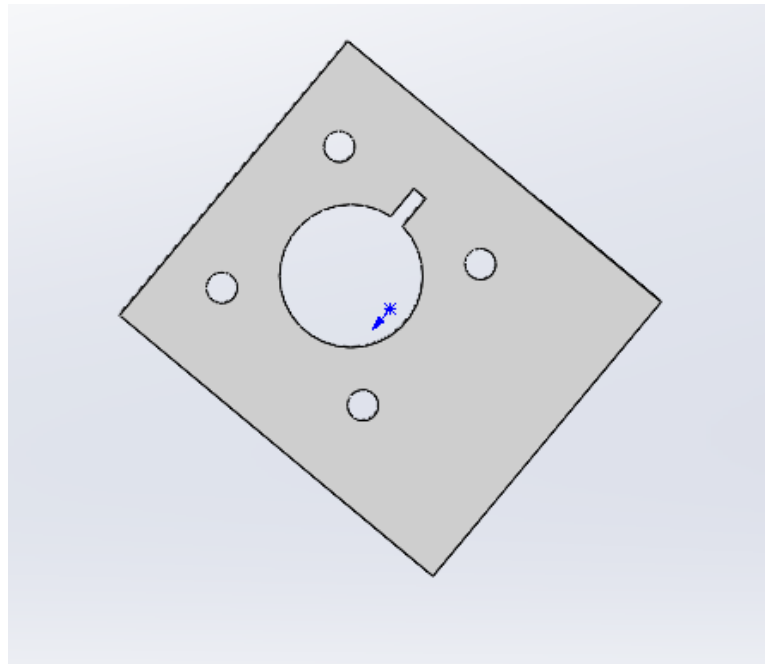
3D Printing Consumables and Time Consumption Data Summary

Part 2: Design one part to be CNC machined:

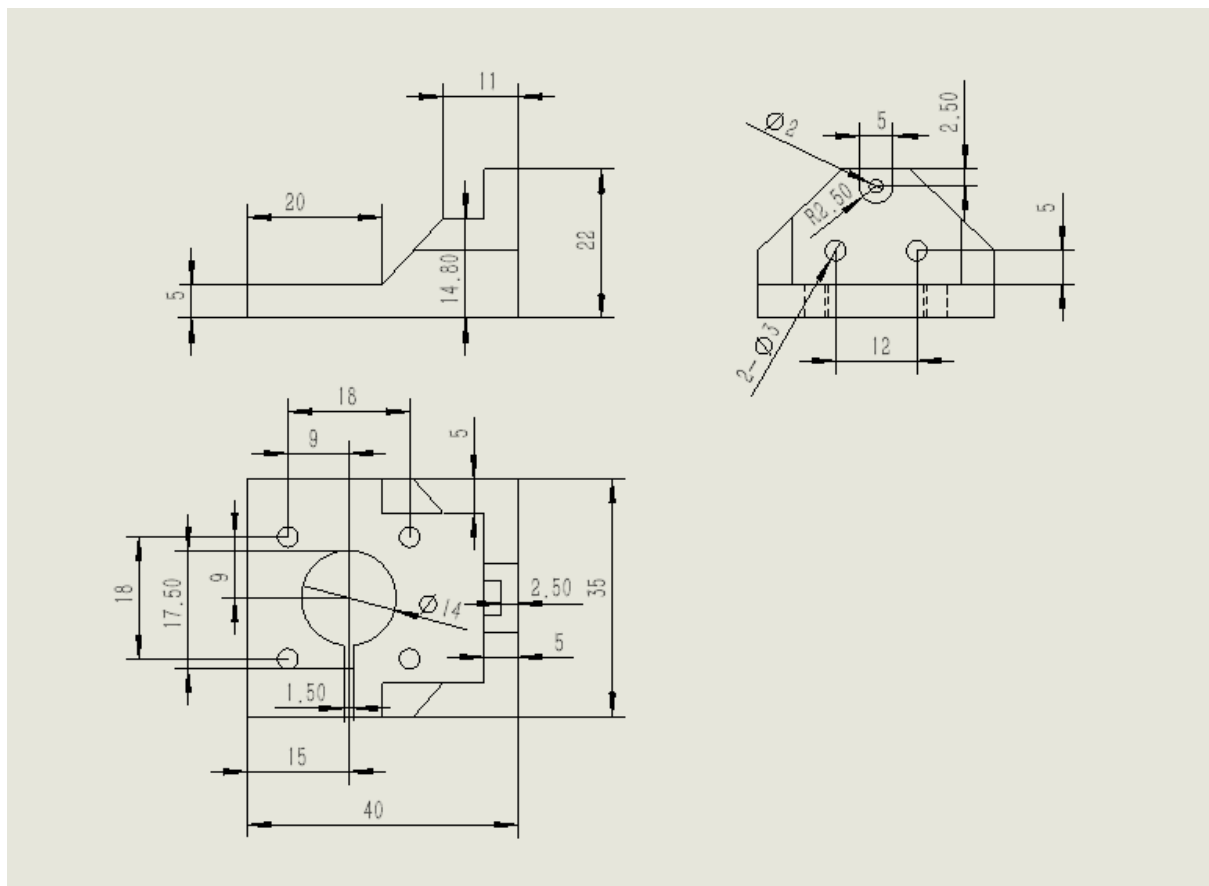
Isometric and orthogonal views for CNC design:



Modelling top view



Elevation view of modelling



2D Engineering Drawing (mm)

Part Function Introduction:

Parts 401 and 402 are symmetrical L-shaped brackets that mount the Prusa printer's Z-axis stepper motors to the frame. One leg attaches vertically to the frame with screws, while the horizontal surface connects to the motor flange, using a rubber pad for vibration damping. Originally 3D printed in PETG, these brackets must provide sufficient rigidity to support the motor weight and resist forces from Z-axis motion.

Design changes:

The redesign of the 401/402 bracket optimises materials, construction and workmanship. The original honeycomb filling is replaced by a solid structure with weight-reducing grooves to avoid closed cavities and improve machinability. Inside corners are rounded to reduce stress, all holes are designed as through holes, and the profile is simplified to minimise tool changes and multi-axis operations, ensuring efficient machining.

Safety Factor Calculation :

The yield strength of 6061-T6 aluminium alloy is approximately 270 MPa. The safety factor is Yield strength divided by maximum stress, So N should be 818 ($270\text{MPa} / 0.33\text{MPa}$). A safety factor of several hundred means that the strength margin of the support is so large that it does not become a limiting factor at all. Even if extreme cases are considered, the safety factor is still rich.

Manufacturing Time and Cost Estimates:

CNC-machined 401/402 supports are suitable for medium-sized series production and are very economical. Each piece is machined from 6061 aluminium alloy billet, weighing approximately 0.068 kg, with a material cost of approximately A\$0.41 and a machining time of approximately 4 minutes. The total cost per hour is estimated at A\$75, the machining cost is about A\$5.00, the total cost per piece is A\$5.41 and the cost per pair is about A\$10.80.

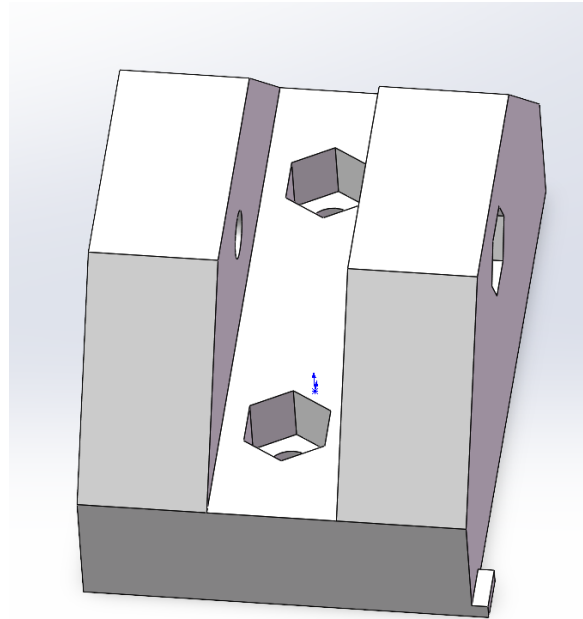
Why use the CNC process

CNC machined 401/402 brackets are chosen for their high rigidity and precision requirements. Compared with plastic parts, CNC aluminium alloy structure is stronger and more precise, avoiding deformation and creep; compared with injection moulding and sheet metal, no mould is required, which is suitable for medium volume production and flexible design adjustment. 6061-T6 material combines light weight and strength, enhancing the stability and life of the

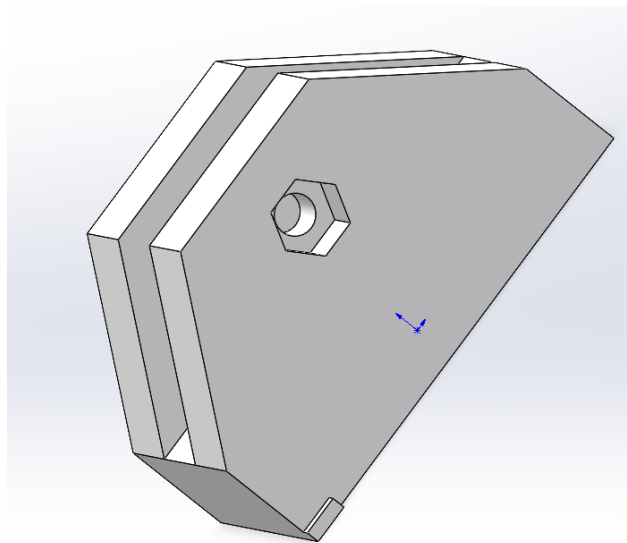
whole machine. Combining performance, cost and adaptability, CNC is the optimal solution for manufacturing such key components.

Part 3 Injection Moulding

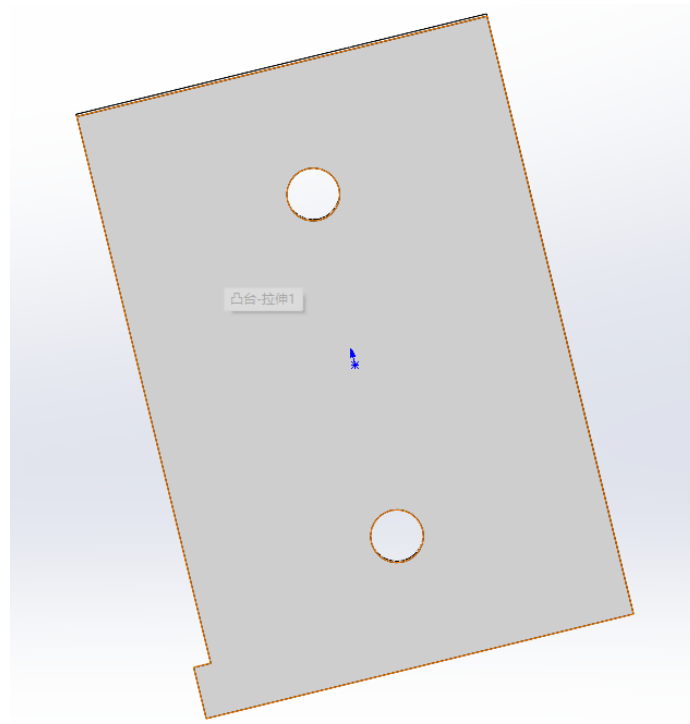
Dimensional and structural dimensioning of injection moulding process-guided parts:



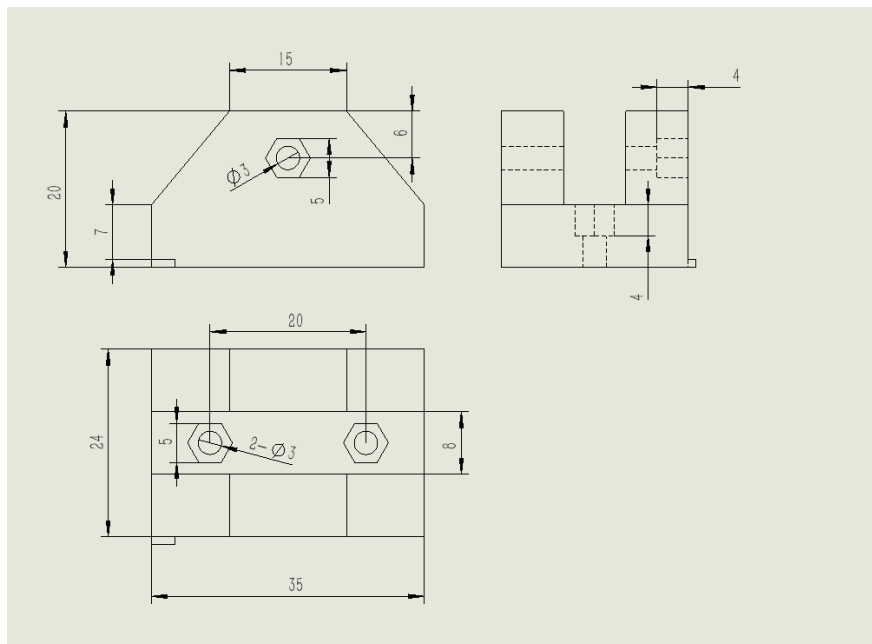
Functional structure part modelling diagram 1



Functional structure part modelling diagram 2



Functional structure part modelling diagram 3



2D Engineering Drawing (mm)

Part Function Introduction:

The 205Y is a bracket assembly for the Prusa i3 printer that holds the idler pulley and adjusts belt tension and is mounted on the front of the frame. The original is 3D printed in PETG

material, weighs approximately 10 grams, and is constructed with mounting holes, bearing slots, and nut mounts to ensure smooth operation of the Y-axis.

Design changes:

Changing the 205Y bracket to injection moulding requires optimising the structure, including increasing the demoulding angle by 2° - 3° , unifying the wall thickness to 3-4mm, adding reinforcement and preventing deformation; changing the sharp corners to rounded corners to improve fluidity; placing the nut holder on the parting surface, which can be embedded with metal parts; slightly enlarging the bearing holes to compensate for shrinkage, and avoiding the inverted buckling of the overall structure to improve injection efficiency and stability.

Safety Factor Calculation:

The 205Y bracket is injection moulded in ABS, with a yield strength of about 40 MPa and an actual stress of only 0.67 MPa, with a static safety factor of 60, far exceeding engineering requirements. Compared to FDM parts, injection moulded parts are isotropic, have no interlayer weakness and are more reliable.

Manufacturing Time and Cost Estimates:

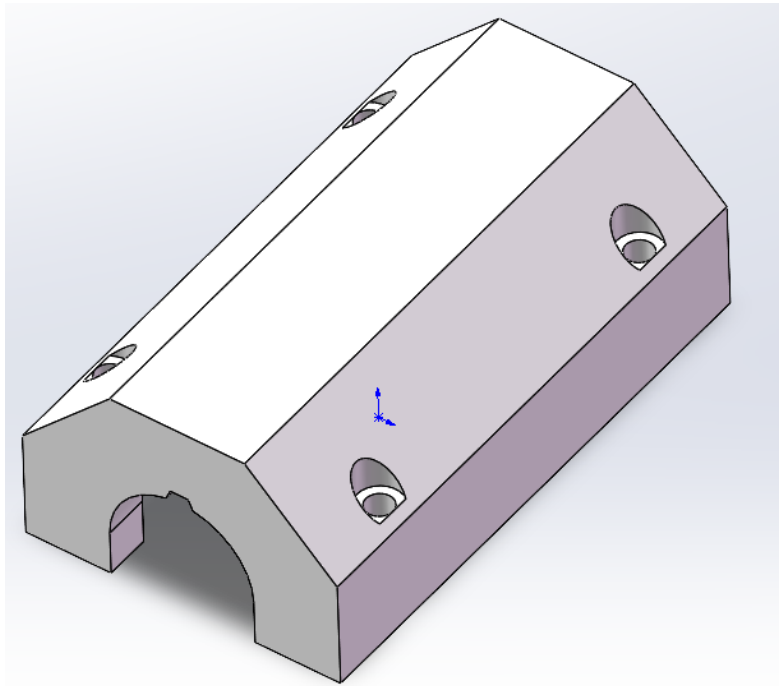
Switching the 205Y bracket to injection moulding greatly reduces cost and boosts efficiency. Material cost is A\$0.03, cycle time is 15 seconds, and hourly output reaches 240 units. With A\$175/hour in equipment and labour, processing cost is about A\$0.73 per piece. Adding mould amortisation (A\$0.15–A\$0.30), the total cost is around A\$1.0, potentially dropping below A\$0.5 with scale—far more economical than 3D printing.

Why use the Injection Moulding process

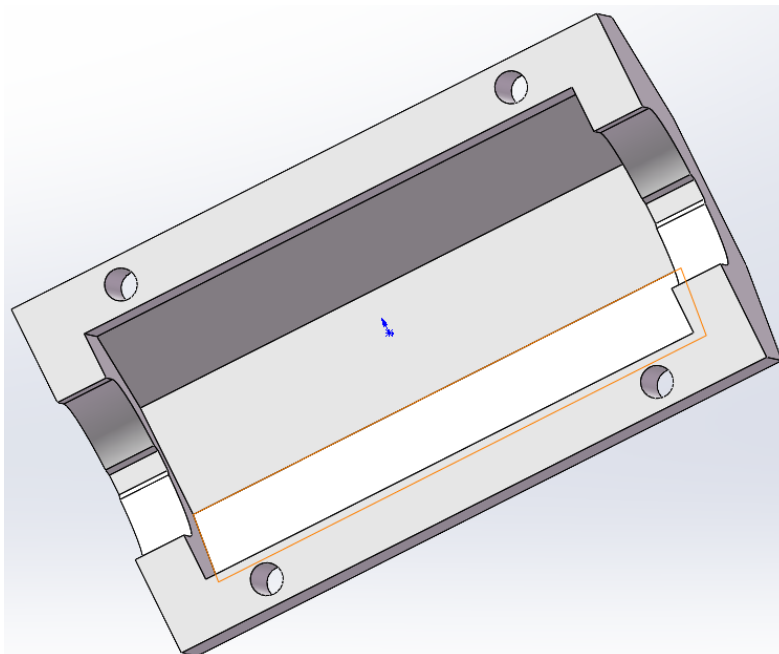
Injection moulding was chosen as the mass production process for the 205Y bracket due to its high efficiency, low cost and consistent quality. Compared to 3D printing, injection moulding is more productive and cheaper per piece, making it suitable for mass production. The ABS injection moulded parts have high dimensional accuracy, good strength, can withstand belt tension, have smooth surfaces, and support structural integration and design optimisation, making it an ideal solution for the high-performance, low-cost manufacturing of the 205Y bracket.

Part 4 Sheet metal fabrication

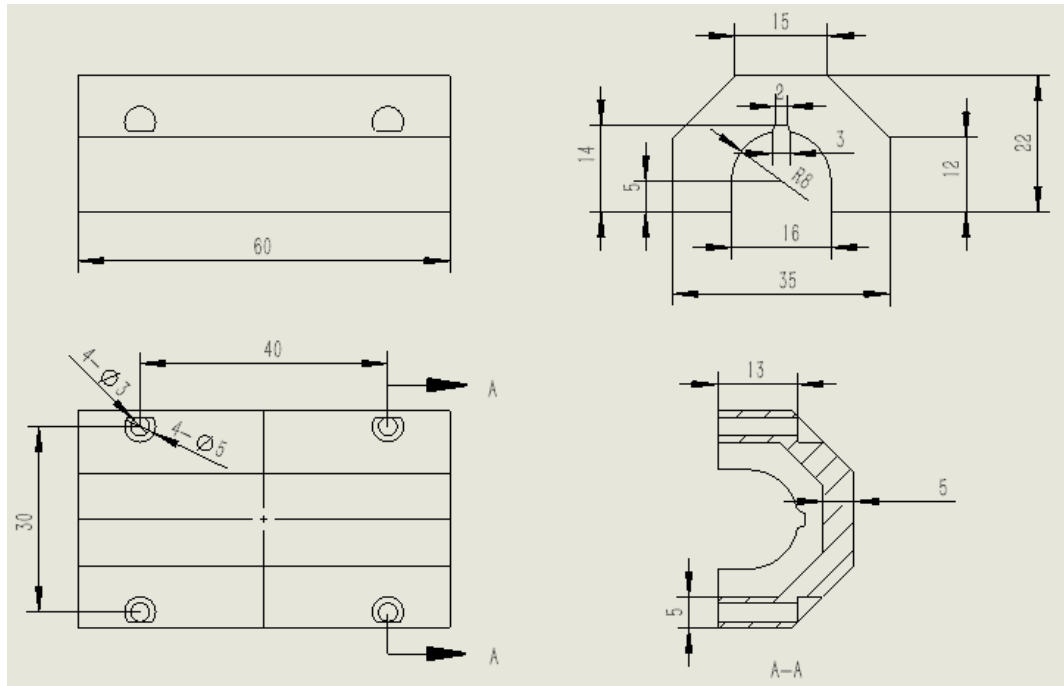
Isometric sketches and engineering drawings for sheet metal working guides



Modelling top view



Elevation view of modelling



2D Engineering Drawing

Part Function Introduction:

The 304X is the ‘X End Bearing Clip’ of the X-axis assembly for Prusa 3D printers. It is originally a PETG 3D printed part and is used in a single unit of 2 pieces, each of which is about 8 grams. Its function is to clamp the two LM10UU bearings into the X-end seat at the end of the X-axis (motor end and idler end) via screws, ensuring smooth movement of the X-axis in the Z-direction.

Design changes:

The redesigned 304X uses 1.0-1.5 mm 304 stainless steel in a simple U-bend configuration with screw holes. Clamping is based on the elasticity of the metal, and with optional pads to increase grip, the new design is suitable for mass production while ensuring performance.

Safety Factor Calculation:

304X clamps made from 1.0–1.2 mm 304 stainless steel provide a safety factor of ~2, exceeding basic requirements. In typical use, lower stress raises the factor above 4. Steel’s fatigue resistance and durability ensure reliability. Strength can be further improved by increasing thickness, using high-strength steel, or adding flanged edges.

Manufacturing Time and Cost Estimates:

Converting 304X clamps to sheet metal process can significantly reduce cost and increase efficiency. The material cost per piece is about A\$0.09, processing time is about 15 seconds, processing cost is about A\$0.625 based on A\$125 cost per hour, and the total cost per piece is about A\$0.72. There is no need for tooling in medium volume, which means the starting cost is low, and it can be optimised further for large volume. Compared to FDM printing which costs over A\$6 per piece, the sheet metal solution has obvious advantages in terms of speed and cost and is the preferred process for mass production of 304X brackets.

Why use the Sheet metal process

The 304X Bearing Clamp has moved from 3D printing to sheet metal processing (laser cutting + bending) due to its simple U-shaped design, high efficiency and low cost. It is easily formed from flat sheet metal with only two bends, avoiding CNC scrap and tooling costs. The sheet metal offers greater strength, precision and durability, clamping stability and ease of assembly. Weighing only 20 grams, it meets structural requirements without adding weight, making it ideal for mass production.